

Department of Engineering Cybernetics  
TTK4215 Adaptive Control  
Solution 1

**Problem 1**

- a) Starting in the early 1950s, the design of autopilots for high-performance aircraft was one of the primary motivations for research activity in adaptive control.
- b) In robust control, constant gain feedback controllers are designed to cope with uncertainties that are within certain bounds. Adaptive control, however, adjust the gains online using parameter estimators to cope with unknown/uncertain dynamics or changing conditions.
- c) One of the main disadvantages of gain scheduling is that the adjustment mechanism of the controller gains is precomputed off-line, and therefore provides no feedback to correct for incorrect schedules or unpredictable changes in the plant dynamics.
- d) In indirect adaptive control, the plant parameters are estimated on-line and used to calculate the controller parameters. In direct adaptive control, the controller gains are estimated directly without any intermediate calculations involving plant parameter estimates. See Figure 1.2 and 1.3 in the book for block diagrams.
- e) An LTI system is minimum-phase if it is stable and its zeros are located in the left-half plane.

- f) In the certainty equivalence approach, the controller design treats the parameter estimates as if they were the true parameters. When implementing the controller, the online parameters estimates are used.
- g) Some basic methods used to design adaptive laws are sensitivity methods (such as the MIT rule), positivity and Lyapunov-based designs, and gradient method and least-squares methods based on estimation error cost criteria.

## Problem 2

Since  $V(t) \geq 0$ ,  $V(t)$  is bounded from below and thus has an infimum given by

$$V_m = \inf_{t \in [0, \infty)} V(t). \quad (1)$$

Therefore, for any  $\epsilon > 0$ , there exists a  $\bar{t} \in [0, \infty)$  such that

$$V_m \leq V(\bar{t}) < V_m + \epsilon. \quad (2)$$

Since  $V(t)$  is non-increasing,  $V(t) \leq V(\bar{t})$  for all  $t \geq \bar{t}$ . It follows that

$$V_m \leq V(t) < V_m + \epsilon, \forall t \geq \bar{t}. \quad (3)$$

Since  $\epsilon$  can be chosen arbitrarily, it follows that

$$\lim_{t \rightarrow \infty} V(t) = V_m. \quad (4)$$

Alternatively, let  $t_0 < t_1 < t_2 < \dots$  such that  $t_k \rightarrow \infty$  and consider the sequence  $V(t_k)$ . Since  $V(t)$  is non-increasing and bounded below by 0,  $V(t_k)$  belongs to the bounded set  $[0, V(0)]$  for all  $t_k$ . Thus, by the Bolzano-Weierstrass theorem there exists a convergent infinite subsequence of  $V(t_k)$ . Since  $V(t_0) \geq V(t_1) \geq V(t_2) \dots$ , elements of  $V(t_k)$  will be squeezed between elements of the subsequence, and so  $V(t_k)$  must converge, too.